



N-Type Solar Panel

USER MANUAL V1.0



Model : CN210W

IEC 61215, IEC 61730

ISO 9001: 2023 Quality management systems

ISO 14001: 2023 Environmental management systems

ISO 45001: 2023 Occupational health and safety management systems

Thank you for purchasing the solar panel. Before using this product, please carefully read this manual. Once you use this product, it will be deemed that you fully acknowledge and accept all the contents of this manual. Please strictly follow the instructions in this manual when using this product and pay close attention to the precautions.

We are not responsible for any direct or indirect losses or injuries caused by improper use of the solar panel, including but not limited to short-circuiting the product, overheating, inserting foreign objects, disassembly, throwing, heavy pressure, and using low-quality controllers, etc.

****Please confirm in advance whether the parameters of this solar panel match the specifications of your equipment. Failure to meet these specifications may result in damage to the products.***

⚠ Product Use Precautions

1. Never attempt to disassemble, inserting foreign objects damage or modify the unit, as this may lead to unforeseen risk.
2. During charging, keep away the solar panel from combustible materials.
3. Avoid dropping, placing heavy objects on, or subjecting this unit to strong impacts.
4. Do not bend the solar panel. Handle with care during use to prevent collisions with the solar panel. Avoid touching it with sharp objects and refrain from forcefully striking the product.
5. Do not step on, stand on, or jump on the surface of the solar panel. Heavy or sharp objects may scratch the solar panel, causing small cracks and affecting output power or posing a risk of injury.
6. Do not pull the power cable hard to avoid the power cable from breaking or falling off.

Product Highlights

 Strengthen Mechanical Load Professional reinforced mechanical loads	 Safety Assurance Under Extreme Heat PV modules have excellent fire resistance
 More Energy Yield Higher energy production makes sure we always achieve your needs	 Anti -PID Excellent Anti-PID performances are ensured by optimized production process and material control

**12V/24V High-Voltage Adaptive**

Featuring a high-voltage architecture, compatible with both 12V and 24V systems for maximum installation flexibility.

**Parallel Connection Design**

Innovative dual-module design ensures shading on one half won't impact the power output of the other.

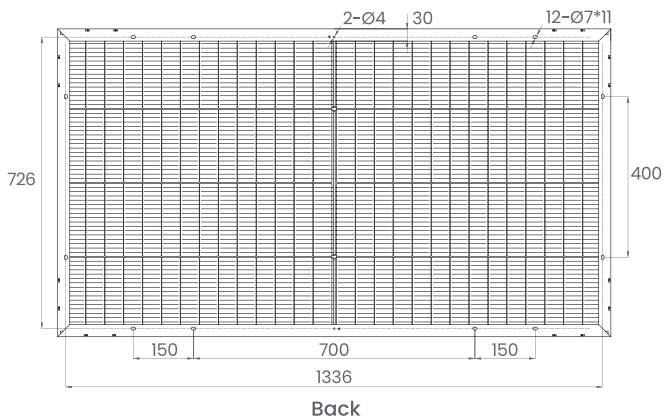
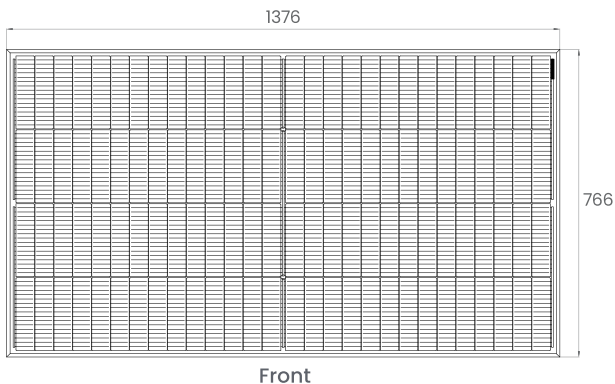
**SMBB Super Multi Main Grid Technology/Quarter-Cell Technology**

SMBB technology creates tougher, more efficient solar modules.

Product Dimensions

Model: CN210W

1376 × 766 × 35 (±5mm)



Mechanical Characteristics(STC)

Model	CN210W
Maximum Power at STC (Pmax)	210W
Maximum Power Voltage at STC (Vmp)	34.2V
Maximum Power Current at STC (Imp)	6.1A
Open-circuit Voltage at STC (Voc)	40.3V
Short-circuit Current at STC (Isc)	6.5A
Cell Type	N-Type, 16BB, 182mm, Monocrystalline Silicon
Cell Efficiency	25%
Power Tolerance	0~5%
Maximum System Voltage	1000V DC
Maximum Series Fuse Rating	25A
First Year Degradation Rate	1%
Linear Degradation Rate	0.4%
Commitment to maintain 84.5% output	25 years

*STC(Standard Test Condition): Irradiance 1000W/m², Cell Temperature 25°C, Air Mass 1.5.

Specifications

Model	CN210W
Dimension	1376 × 766 × 35 (±5 mm) / 54.2 × 30.2 × 1.4 (±0.1 in)
Weight	11.3 kg / 24.9 lb
Glass	3.2mm Tempered Glass
Frame	Anodized Aluminium Alloy
IP Rating	IP68
Max. Wind Load/Snow Load	2400 Pa / 5400 Pa
Connection	Solar Connector

Temperature Coefficients

Model	CN210W
Temperature coefficients of Pmax	-0.30%/K
Temperature coefficients of Voc	-0.25%/K
Temperature coefficients of Isc	0.045%/K
Operating Temperature (°C)	-40°C ~ 85°C

Quick Guide

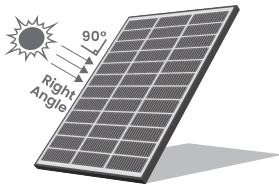
⚠ Key Reminder

1. No power generation at night! Charge power station in daytime.
2. Check the DC input voltage and power range of your power station. Some models have fixed input limits, so even if the solar panel outputs more, it won't charge above that limit. In this case, the power station will still charge normally within the supported range.
3. Please avoid hitting, heavy pressure, dropping, or stepping on the solar panel.

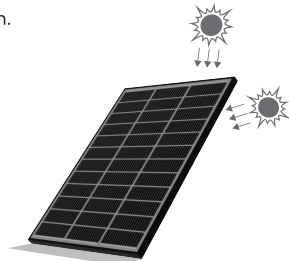
Adjust Panel Angle

★ Face Sunlight Directly!

Adjust panel angle to PERPENDICULAR to the sun.



✓ RIGHT

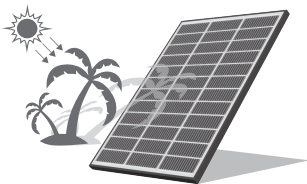


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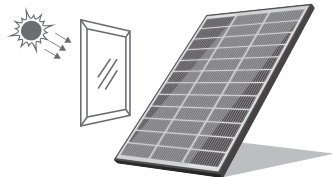
Avoid Obstructions

Avoid Using Solar Panels in Areas with Obstructions.

Not Recommended to Use Solar Panels Behind Glass/Windows/Curtains/Screens.



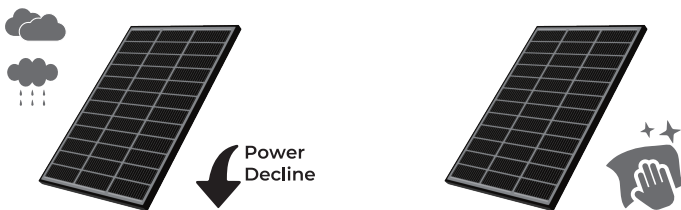
✗ WRONG



✗ WRONG

Power Loss Factors

1. In times of low sunlight (sunlight obscured by clouds, or rainy days), the power generation of solar panels will be reduced.
2. Dirt on the surface of solar panels can reduce power generation. Please clean its surface regularly.



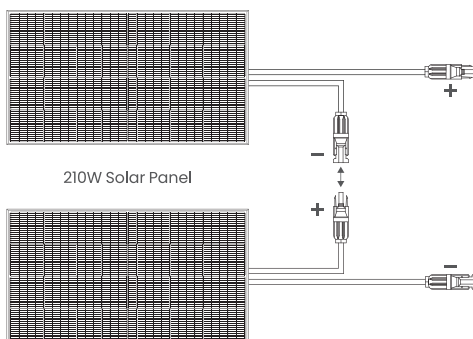
Precautions for Solar Panel Series and Parallel Connections

1. For best results, always use solar panels of the same brand and specifications to ensure consistent electrical performance and maximize system efficiency and stability.
2. Connection Methods:
 - Series Connection: Current remains the same while voltages add up; the system current is limited by the panel with the lowest current.
 - Parallel Connection: Voltage remains the same while currents add up; the system voltage is limited by the panel with the lowest voltage. (When combining panels in a series-parallel configuration, always check controller limits to ensure compatibility and safety.)
3. If you are using a PWM controller, please note that this product is only compatible with 24V PWM system. Connecting it to a 12V PWM controller will result in significant power loss. When building a 12V system, it's highly recommended to use a MPPT controller.

If you need videos on series or parallel connection of solar panels, you can contact us at: callsun-service@outlook.com

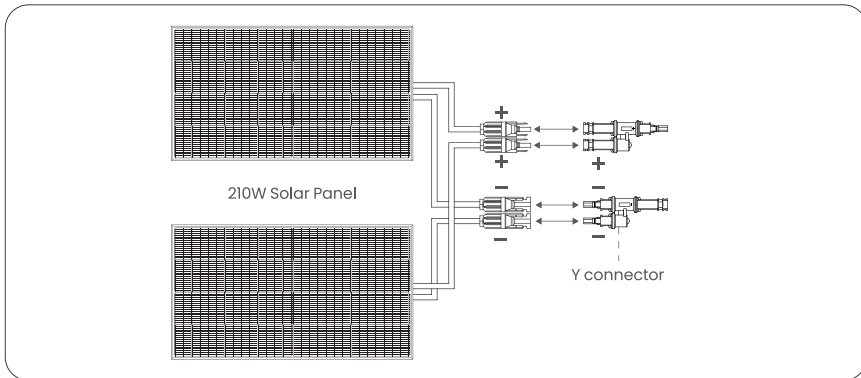
Connect in Series

Connect the positive terminal of one solar panel to the negative terminal of the other solar panel.



Connect in Parallel

Connect the positive terminals of both solar panels to the same side of the corresponding Y connector. Then, connect the negative terminals of both panels to the same side of the other corresponding Y connector.



Product Maintenance

After long-term use, dust may adhere to the surface of the panel, blocking sunlight and reducing the cells ability to absorb sunlight. Regularly clean the solar panel by gently wiping it with a damp cloth. Do not use corrosive liquids to clean the solar panel.

Suitable Environments

To get the maximum power, the solar panel must be exposed to direct sunlight with no obstructions on its surface. Please use it under clear sunlight without clouds, trees, or buildings blocking the sunlight.

Warranty Policy

We provide a 10-year warranty service for materials and workmanship on this product. Within 10 years from the date of purchase, if the product experiences faults under normal usage, please apply for warranty with the purchase receipt.

The following situations are not covered by the warranty:

1. Cosmetic flaws caused during use.
2. Product malfunctions resulting from self-disassembly, repairs, or human-induced causes.
3. Damages caused by uncontrollable factors such as natural disasters, thunderstorms, accidents, and the like.

Warranty Application Process:

1. Please contact our customer service team, providing the purchase receipt and a detailed description of the issue.
2. Our customer service team will assist you in submitting the necessary information and documents.
3. After confirmation, we will provide you with an appropriate warranty solution.

If you have any questions or need assistance, please feel free to reach out to us via the following email: **callsun-service@outlook.com**

Specifications are subject to change without notice!

Shenzhen Guangchuxinneng Technology Co., Ltd.

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